

Reinforcement Learning to Achieve Real-time Control of a Quadruple Inverted Pendulum

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Abstract: For the first time in model-free control, the real-time swing up and balancing task of a quadruple inverted pendulum (QIP) is successfully performed through reinforcement learning (RL) in a sample-efficient, data-driven manner. As a full state-feedback closed-loop form, the policy network designed in this paper can achieve the control objective of efficiently swinging up and balancing a QIP even without any pre-computed reference trajectories and any mathematical models. To do so, this work first defines the Markov decision process (MDP) for QIP control and then applies a state-of-the-art off-policy RL algorithm, truncated quantile critics (TQC), to accomplish the control objective by using the probabilistic distribution of Q-values. For high sample-efficiency, virtual experience replay (VER) is used to leverage the geometrically symmetric structure and thereby reduce the learning time. The proposed demonstration shows that the complex QIP control problem, difficult-to-solve through classical model-based approaches, can be solved only from measured data without accurate mathematical model derivation and stability-guaranteed control design.

Keywords: Quadruple inverted pendulum, reinforcement learning, truncated quantile critic, virtual experience replay.

1. INTRODUCTION

An inverted pendulum system is a representative example of nonlinear dynamic systems, characterized by instability and non-minimum phase properties. Moreover, it is classified as an underactuated system, where the number of control inputs is less than the system's degrees of freedom, providing significant research topics in control theory. In particular, multi-link inverted pendulum systems consist of multiple interconnected links, resulting in highly nonlinear dynamics with high-dimensional complexity. As shown in [1], the stabilizability radius — a measure of how robustly the system can be stabilized under feedback control — dramatically decreases as the number of pendulum links increases. This means that beyond structural complexity, higher-order inverted pendulums are fundamentally more difficult to stabilize using any control technique.

Given the challenges inherent to multi-link inverted pendulum systems, their control typically involves three primary tasks: swing-up control for transitioning the pen-

dulum from a hanging-down position to an upright position; balancing control for stabilizing the pendulum in the upright position; and side-stepping control for moving the pendulum laterally [2,3]. Among these, swing-up control is widely recognized as the most challenging. A major milestone was achieved in 2007, when Graichen successfully demonstrated swing-up control of a double inverted pendulum using a two-degree-of-freedom (2-DOF) control structure based on feedforward and feedback design [2]. This trajectory optimization approach was later extended to triple inverted pendulums by incorporating additional constraints and adopting numerical methods such as direct collocation [4,5]. While these trajectory-based approaches have shown success, they are fundamentally rooted in classical control paradigms that depend heavily on accurate dynamic models and precomputed reference trajectories. However, for high-dimensional and nonlinear systems such as multi-link inverted pendulums, these mathematical model based classical approaches have practical limitations with open-loop trajectory based strategy and inevitable reality-to-simulation gap. Specif-

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ically, classical control schemes based on reference trajectories are fundamentally open-loop and hence may not work properly if the system deviates from prescribed trajectories. In addition, as the system complexity increases, it becomes really difficult to obtain accurate dynamic models and relevant parameters, which leads to a significant decline in control performance.

To overcome the practical constraints associated with conventional model-based control schemes, reinforcement learning (RL) has emerged as a promising alternative approach [6-10]. RL is a model-free learning-based approach that enables an agent to learn an optimal policy directly from data without relying on explicit, mathematically described system dynamics. In particular, RL can operate reliably even in complex systems with high-dimensional state spaces and nonlinearities [11]. Since the agent learns autonomously through interactions with the environment, RL offers greater design flexibility and broader applicability compared to classical control methods. Above all, RL can develop closed-loop state-feedback control strategies that work well without reference trajectories and steer any starting state to any desired target. It has recently been demonstrated that model-free RL can effectively learn closed-loop trajectory-free optimal control policies even in high-dimensional state spaces, by successfully stabilizing a triple inverted pendulum in real-time without relying on mathematical dynamic models [12].

Meanwhile, recent advances in RL have led to the widespread adoption of model-free, off-policy actor-critic algorithms for a broad range of control tasks [13-15], including soft actor-critic (SAC) [16], truncated quantile critics (TQC) [17], deep deterministic policy gradient (DDPG) [18], and twin delayed DDPG (TD3) [19]. However, the majority of these studies have been conducted on relatively low-dimensional or structurally simple systems, such as 2-DOF manipulators or planar robotic platforms. As a result, the applicability of RL to highly nonlinear, high-dimensional, underactuated systems — such as a quadruple inverted pendulum (QIP) — has remained largely unexplored due to the high system complexity and the inherent challenges in control. Furthermore, existing studies on the QIP system fail to find a connection with RL, by still focusing on classical approaches such as deriving accurate dynamic models, performing linearization-based analyses, and designing primitive balancing controllers [20-22]. Moreover, these works were only interested in achieving balance through modeling and linearization without a swing-up process, which clearly shows that existing model-based control schemes are insufficient to handle the complex QIP system. To the best of our knowledge, no prior study has successfully achieved swing-up control of the QIP system, nor has there been any research attempting it without relying on a priori dynamic information. Accordingly, this paper

aims to achieve fully model-free swing-up control of the QIP system through RL. Specifically we integrated truncated quantile critics (TQC) [17], an RL algorithm effective for policy optimization in continuous action spaces, with the virtual experience replay (VER) technique proposed in [12]. TQC mitigates overestimation bias by truncating the upper quantile estimates in the value distribution, while VER enhances training efficiency by leveraging virtually generated data. The combination of these two techniques enables stable and sample-efficient learning in high-dimensional continuous control environments, making it particularly suitable for complex systems such as the QIP. This paper is the first to demonstrate that the control of swing-up and balancing of the QIP system can be effectively achieved by employing a full state-feedback, closed-loop RL policy, without relying on mathematical models or predefined reference trajectories. It is believed that this achievement marks an important milestone in the history of control engineering. A supplementary video clip is provided for further understanding [23].

More specifically, the proposed QIP control is

- 1) Designed to perform properly in all states without prescribing reference trajectories.
- 2) Of a tractable form for extending to multi-task and multi-modal learning.
- 3) Suitable for enhancing robustness through domain randomization.

The remainder of this paper is organized as follows: Section 2 introduces the structure and characteristics of the QIP system. Section 3 discusses the design of the RL-based controller for swinging up a QIP, along with details of the training environment. Section 4 presents the training results and validation through experiments conducted in a simulated setting. Section 5 offers a discussion on the robustness of the trained controller under structural parameter variations, as well as computational considerations for scaling the approach to more complex systems. Finally, Section 6 concludes the study.

2. QUADRUPLE INVERTED PENDULUM SYSTEM

Fig. 1 illustrates the mechanical concept of the QIP system. The QIP system consists of a cart and four serially connected links. The cart can move horizontally when an external force F is applied, while all four links are connected through passive joints.

Fig. 1 shows the physical parameters of the QIP system. For each link i ($i = 1, \dots, 4$),

- 1) m_i represents the mass of the i -th link,
- 2) L_i is the length of the i -th link,
- 3) l_i denotes the distance from the i -th joint to the center of mass of the i -th link,

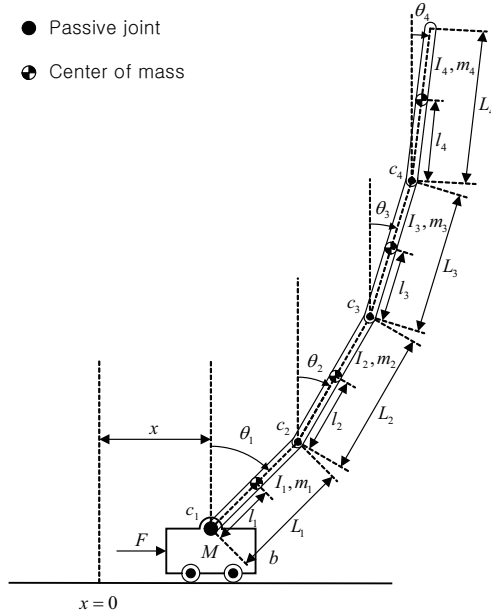


Fig. 1. A schematic view of a quadruple inverted pendulum system.

- 4) I_i is the moment of inertia of the i -th link, and
- 5) c_i represents the friction coefficient at the i -th joint.

Additionally, M denotes the mass of the cart, and b represents the friction coefficient of the cart. The applied force on the cart is denoted as F , while x represents the cart's displacement, and θ_i ($i = 1, \dots, 4$) indicates the rotational displacement of each link.

The dynamic equations of the QIP system can be derived using the Euler-Lagrange equation (1). The generalized coordinate $q = [x, \theta_1, \theta_2, \theta_3, \theta_4]^T$ represents the vector of the system's degrees of freedom, while the generalized force $\tau = [F, 0, 0, 0, 0]^T$ corresponds to the external forces or torques applied to each generalized coordinate. The mathematical model of the QIP system can be described as

$$D(q)\ddot{q} + C(q, \dot{q})\dot{q} + G\dot{q} + g(q) = \tau, \quad (1)$$

where $D(q)$ represents the QIP system's inertia matrix, $C(q, \dot{q})$ is the Coriolis matrix, G denotes the damping vector, and $g(q)$ corresponds to the gravitational vector. Since $D(q)$ is symmetric and configuration-dependent, its structure plays a crucial role in the QIP system's dynamics and is given as follows:

$$D(q) = \begin{bmatrix} D_{11} & D_{12} & D_{13} & D_{14} & D_{15} \\ \star & D_{22} & D_{23} & D_{24} & D_{25} \\ \star & \star & D_{33} & D_{34} & D_{35} \\ \star & \star & \star & D_{44} & D_{45} \\ \star & \star & \star & \star & D_{55} \end{bmatrix}, \quad (2)$$

where $\star$ represents the symmetric elements of the matrix, and the elements of $D(q)$ are defined as follows:

$$\begin{aligned} D_{11} &= a_0, \\ D_{12} &= a_1 \cos \theta_1, \\ D_{13} &= a_2 \cos \theta_2, \\ D_{14} &= a_3 \cos \theta_3, \\ D_{15} &= a_4 \cos \theta_4, \\ D_{22} &= b_1, \\ D_{23} &= a_2 l_1 \cos(\theta_1 - \theta_2), \\ D_{24} &= a_3 l_1 \cos(\theta_1 - \theta_3), \\ D_{25} &= a_4 l_1 \cos(\theta_1 - \theta_4), \\ D_{33} &= b_2, \\ D_{34} &= a_3 l_2 \cos(\theta_2 - \theta_3), \\ D_{35} &= a_4 l_2 \cos(\theta_2 - \theta_4), \\ D_{44} &= b_3, \\ D_{45} &= a_4 l_3 \cos(\theta_3 - \theta_4), \\ D_{55} &= b_4. \end{aligned}$$

The coefficients a_0 , a_i , and b_i ($i = 1, \dots, 4$) are further defined as

$$\begin{aligned} a_0 &= M + \sum_{i=1}^4 m_i, \\ a_i &= m_i l_i + \sum_{j=i+1}^4 m_j L_j, \\ b_i &= I_i + m_i l_i^2 + \sum_{j=i+1}^4 m_j L_j^2. \end{aligned}$$

As seen from (1) and (2), the QIP system exhibits high-dimensionality and complexity, where each link is dynamically coupled with the others. Consequently, the inertia matrix $D(q)$, the Coriolis matrix $C(q, \dot{q})$, and the gravitational vector $g(q)$ change nonlinearly as functions of q and $\dot{q}$. Additionally, the only control input is the external force F applied to the cart, while the remaining links are passively controlled through their joints. This high-dimensionality and nonlinearity make it challenging to design an effective controller using classical control techniques. Furthermore, they highlight the necessity of nonlinear model-free approaches such as RL, which can create control policies directly through training without requiring an explicit mathematical representation of the complex system model.

3. CONTROLLER DESIGN

In this section, we discuss the controller design for performing swing-up control of the QIP using RL. First, the swing-up problem of the QIP is formulated as a Markov decision process (MDP) that consists of a 5-tuple, $(\mathcal{S}, \mathcal{A}, p, r, \gamma)$, where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, $p: \mathcal{S} \times \mathcal{S} \times \mathcal{A} \rightarrow [0, \infty)$ is the state-transition probability density function, $r: \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ is the reward function, and

$\gamma \in [0, 1]$ is the discount factor. Then, the TQC algorithm is employed to design the controller, with an additional implementation of VER to leverage the system's symmetry. Finally, we present the training environment and its related components using our custom-designed TQC algorithm.

3.1. MDP formulation

3.1.1 State space ($\mathcal{S}$)

The state space $\mathcal{S}$ is a set of all possible states of the QIP system. Its state variable consists of the angles and angular velocities of four links, as well as the displacement and velocity of the cart. Mathematically, it is expressed as follows:

$$s_t = [x_t, \dot{x}_t, \sin(\theta_{1,t}), \cos(\theta_{1,t}), \dot{\theta}_{1,t}, \dots, \sin(\theta_{4,t}), \cos(\theta_{4,t}), \dot{\theta}_{4,t}], \quad (3)$$

where t represents the time step, s_t denotes the system state at time t , x_t indicates the position of the cart, and $\theta_{i,t}$ represents the angle of each link. x_t and $\theta_{i,t}$ are shown in Fig. 1.

In particular, we employ the sine and cosine representation for each joint angle to preserve the periodicity of angular variables and avoid discontinuities around the $-\pi$ and π boundaries. This transformation not only ensures continuity and smooth gradients for policy learning, but also serves as a form of normalization, facilitating the generalization across the state space. In our case, it proves crucial for enabling the agent to learn robust swing-up behavior in the high-dimensional QIP system.

3.1.2 Action space ($\mathcal{A}$)

The action space $\mathcal{A}$ is a set of all feasible force F applied to the cart and is set as a continuous value. Additionally, to better reflect real-world system constraints, the control input is restricted to $a \in [-40\text{N}, 40\text{N}]$.

3.1.3 Reward function ($r(\cdot, \cdot)$)

The swing-up control of the QIP system refers to the process of transitioning the system from its initial state, where all links are hanging downward, defined as $x_0 = 0$, $\dot{x}_0 = 0$, $\theta_{i,0} = \pi$, $\dot{\theta}_{i,0} = 0$ ($i = 1, \dots, 4$), to the target state, where all links are in the upright position, given by $x^* = 0$, $\dot{x}^* = 0$, $\theta_i^* = 0$, $\dot{\theta}_i^* = 0$ ($i = 1, \dots, 4$). The reward function $r(s, a)$ is designed to reflect this control objective, and to achieve swing-up control. The reward function is determined as follows:

$$r(s, a) = r_{\text{act}}(a) \times r_{\text{pos}}(x) \times r_{\text{angle}}(\theta) \times r_{\text{vel}}(\dot{\theta}), \quad (4)$$

where $r_{\text{act}}(a)$, $r_{\text{pos}}(x)$, $r_{\text{angle}}(\theta)$, and $r_{\text{vel}}(\dot{\theta})$ are given by

$$\begin{aligned} r_{\text{act}}(a) &= 0.8 + 0.2 \max(1 - a^2, 0), \\ r_{\text{pos}}(x) &= 0.5 + 0.5 \exp(-0.7x^2), \end{aligned}$$

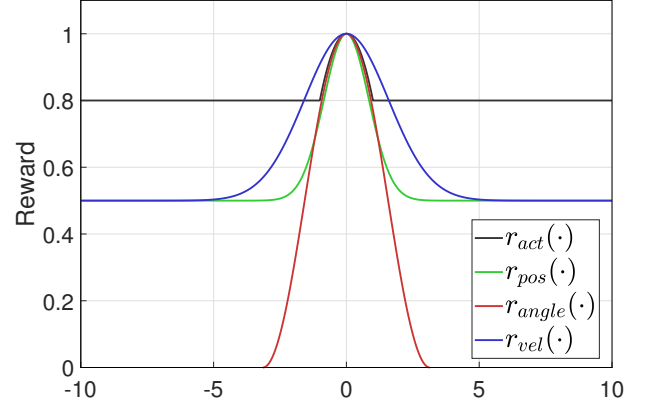


Fig. 2. Four component functions for the reward.

$$r_{\text{angle}}(\theta) = \prod_{i=1}^4 (0.5 + 0.5 \cos \theta_i),$$

$$r_{\text{vel}}(\dot{\theta}) = \min(0.5 + 0.5 \exp(-0.2 \cdot \dot{\theta}_i^2)), \quad i \in \{1, \dots, 4\}.$$

Four components for the total reward function are shown in Fig. 2. The reward function is carefully designed to reflect multiple control objectives such as upright stabilization, motion smoothness, and energy efficiency. Specifically, the exponential penalties apply to velocity and position error terms, guiding the agent toward smooth and centered control behaviors, while the multiplicative structure of angle-related cosine terms emphasizes simultaneous alignment of all pendulum links. The coefficients and exponential slopes are selected empirically based on the observation that excessively steep penalties often hinder convergence in high-dimensional spaces and smooth gradients enable stable and consistent learning. The total reward function is formulated as the product of all individual reward terms, thereby integrating multiple control objectives into a unified cost metric.

3.2. Implementation details

3.2.1 Truncated quantile critics with virtual replay buffer

Virtual experience replay (VER) [12] is a useful technique designed to improve the efficiency of experience replay when applying off-policy reinforcement learning algorithms to geometrically symmetric systems. It focuses on generating virtual experiences by exploiting the geometric structures of symmetrical systems such as the QIP system. This method transforms collected data from the existing experience replay buffer to generate new virtual data, which is then utilized in training. In this paper, this technique is combined with the off-policy reinforcement learning algorithm, TQC [17], to design the controller. TQC is an algorithm designed to mitigate the overestimation bias by integrating the strengths of soft actor-critic (SAC) [16] and quantile regression deep Q-network (QR-DQN) [24]. This algorithm introduces the concept of quantile truncation, which eliminates overestimated quantiles, enabling a

more accurate and reliable distribution estimation.

The critic network of TQC learns the Q-value distribution $Z_{\psi_n}(s, a)$ for the state-action pair (s, a) and is trained to minimize the Huber quantile loss based on the target quantile $y_i(s, a)$. The Huber quantile loss $\rho_\tau^H(\cdot)$ follows the same formulation as in [24] and is given by

$$\rho_\tau^H(u) = |\tau - \mathbb{I}(u < 0)| \mathcal{L}_H^1(u), \quad (5)$$

where $\mathcal{L}_H^1(\cdot)$ is defined by

$$\mathcal{L}_H^1(u) = \begin{cases} \frac{1}{2}u^2, & \text{if } |u| \leq 1, \\ |u| - \frac{1}{2}, & \text{otherwise.} \end{cases}$$

Letting $z_i(s, a)$ be the i -th smallest atom among the kN ones from N critics. It follows then that the loss function for training the n -th critic can be taken to be as follows:

$$\mathbb{E}_{(s, a, r, s') \sim D^+, a' \sim \pi_\phi(\cdot|s')} \left[\frac{1}{kNM} \sum_{m=1}^M \sum_{i=1}^{kN} \rho_{\tau_m}^H(y_i(s, a) - \theta_{\psi_n}^m(s, a)) \right],$$

where $y_i(s, a)$ is the ϕ -parameterized policy and $\pi_\phi(\cdot|s)$ is given by

$$y_i(s, a) = r(s, a) + \gamma [z_i(s', a') - \alpha \log \pi_\phi(a'|s')]. \quad (6)$$

Here, N represents the number of critic networks, M denotes the number of atoms in each critic, and the kN smallest atoms among NM ones are selected to obtain the target distribution. The replay buffer D^+ contains both real sampled data and virtual experience data extended through VER, and it is expressed as $D^+ = \{s_t, a_t, r_t, s_{t+1}, \tilde{s}_t, \tilde{a}_t, r_t, \tilde{s}_{t+1}\}$. The extended virtual experience data is derived based on the symmetry of the QIP system, following the same approach as in [12], where $\tilde{s}_t = -s_t$, $\tilde{a}_t = -a_t$, and $\tilde{s}_{t+1} = -s_{t+1}$. Such data augmentation plays a huge role in improving sampling efficiency. Additionally, $y_i(s, a)$, $i = 1, \dots, kN$, represents the atoms of the target distribution, which reflects the current reward and the distribution of future Q-values, while $\theta_{\psi_n}^m(s, a)$ denotes the m -th quantile value returned by the n -th critic network. And also, s' denotes the next state sampled from the environment given the current state-action pair (s, a) , and a' is the action sampled from the policy $\pi_\phi(\cdot|s')$.

The actor network learns the policy $\pi_\phi(a|s)$, which selects an action a given a state s . The actor network is trained to maximize both the expected Q-value returned by the critic networks and the entropy of the policy. The loss function of the actor network is given by

$$\mathbb{E}_{s \sim D^+, a \sim \pi_\phi(\cdot|s)} \left[\alpha \log \pi_\phi(a|s) - \frac{1}{NM} \sum_{m,n=1}^{M,N} \theta_{\psi_n}^m(s, a) \right], \quad (7)$$

Algorithm 1: Training procedure for QIP control.

- 1) **Initialize** policy π_ϕ , critics $Q_{\psi_1}, \dots, Q_{\psi_N}$, replay buffer $\mathcal{D}$
 - 2) Set truncation level d , learning rate α , discount factor γ
 - 3) **for** each episode **do**
 - 4) Reset environment and observe s_0
 - 5) **for** each time step t **do**
 - 6) Select $a_t \sim \pi_\phi(\cdot|s_t)$ with exploration
 - 7) Execute a_t , observe s_{t+1} and reward r_t
 - 8) Construct **VER**
 - 9) $\mathcal{D} \leftarrow \mathcal{D} \cup \{(s_t, a_t, r_t, s_{t+1}), (\tilde{s}_t, \tilde{a}_t, r_t, \tilde{s}_{t+1})\}$
 - 10) **if** $\text{len}(\mathcal{D}) > \text{batch size}$ **then**
 - 11) Sample minibatch $\{(s, a, r, s')\} \sim \mathcal{D}$
 - 12) Estimate targets y_i
 - 13) Truncate top- d from total NM quantiles
 - 14) Update critic network ψ_n
 - 15) Update actor network ϕ
 - 16) **end if**
 - 17) $s_t \leftarrow s_{t+1}$
 - 18) **end for**
 - 19) **end for**
-

where α is the weighting parameter. It is noted that the full atoms are employed for updating the policy network as in the original TQC algorithm [17].

To provide a clearer overview of the learning framework used in this study, the entire training procedure is described in Algorithm 1. This algorithm outlines how the TQC-based policy is trained using quantile value estimates and VER, incorporating critical components such as target estimation, quantile truncation, and entropy-regularized actor updates.

3.2.2 Learning setup

The simulation model of the QIP was provided in the form of Python code by Koh Young Technology, Inc. and was used in combination with the TQC algorithm for training and evaluating. The agent interacts with the environment at a sampling time of 20 ms, and each episode consists of 1000 time steps.

The mechanical parameters of the QIP system were shown in Table 1, and the hyperparameters used for train-

Table 1. Mechanical parameters of the quadruple pendulum on a cart.

Parameter	Cart	Link 1	Link 2	Link 3	Link 4
Mass (kg)	1.000	0.130	0.149	0.168	0.172
Length (m)	–	0.191	0.227	0.263	0.307
CoG dist (m)	–	0.110	0.128	0.146	0.154
MoI (kg·m ²)	–	0.00058	0.00090	0.00132	0.00158

*CoG: Center of gravity, **MoI: Momentum of inertia

Table 2. Hyperparameters involved in the TQC algorithm.

Hyperparameter	Value
Optimizer	Adam [25]
Learning rate	0.0003
Discount factor (γ)	0.99
Replay buffer size	1e6
Number of critics (N)	5
Number of hidden layers in the critic networks	3
Size of hidden layers in the critic networks	512
Number of hidden layers in the policy networks	3
Size of hidden layers in the policy networks	512
Minibatch size	256
Activation function	ReLU
Target smoothing coefficient (β)	0.005
Number of atoms (M)	25
Number of truncated quantiles (d)	4

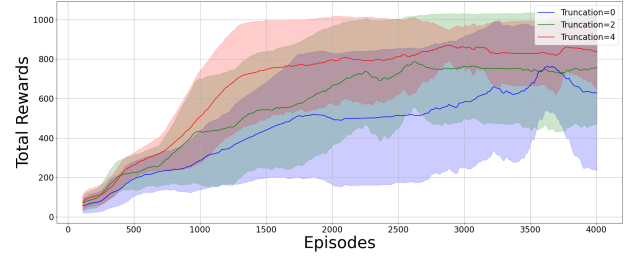
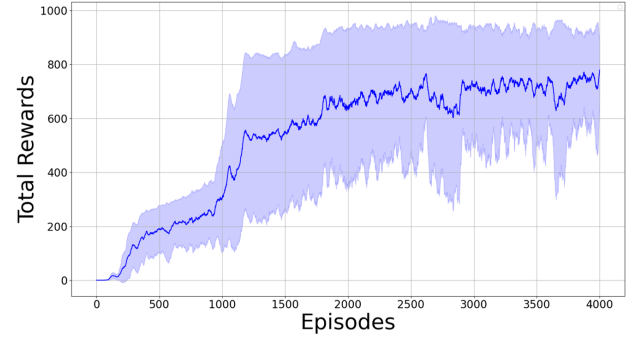
ing were listed in Table 2. The policy network was designed with three hidden layers of 512 neurons each, a structure that was empirically tuned to balance learning capacity and training stability. The number of critic networks ($N = 5$) was selected based on the findings in [17], which demonstrated that performance tended to saturate beyond a certain number of critics. In addition, the number of atoms ($M = 25$) was adopted directly from [17] to maintain consistency with prior studies.

To determine the optimal quantile truncation level in the TQC framework, we conducted systematic ablation experiments in the QIP environment. As depicted in Fig. 3, we tested various truncation levels ($d = 0, 2, 4$) by dropping the top d quantiles from the aggregated NM quantile mixture distribution. The results showed that truncating the highest 4 quantiles (i.e., retaining the lowest $kN = 121$ quantiles from the total $N \times M = 125$) consistently yielded the highest performance and fastest convergence. This implies that selecting an appropriate truncation threshold played a critical role in mitigating overestimation bias without excessively narrowing the expressive power of the value distribution.

All remaining hyperparameters followed the default settings used in [17].

4. EXPERIMENTAL RESULT

The final training results are shown in Fig. 4. Over 4 million time steps, the average reward gradually increases and eventually settles down, which means that the training process of the designed policy network is performed successfully.

**Fig. 3.** Performance comparison of different quantile truncation levels in the QIP environment.**Fig. 4.** Learning curves over 10 random seeds. The shaded region represents the standard deviation, and the bold line indicates the mean of total rewards.

The trained policy neural network was validated under the simulation environment provided by Koh Young Technology, Inc., and its evaluation was performed with a sampling time of 20 ms. Fig. 5 illustrates trajectories of state variables and control action generated when performing swing-up control with the trained policy neural network. To complement the numerical state trajectories shown in Fig. 5, Fig. 6 provides visual snapshots of the QIP system's dynamic motion over time, offering an intuitive depiction of the swing-up process. The control system successfully reached the stable region ($-0.05 \leq \theta_i \leq 0.05$, $-0.05 \leq \dot{\theta}_i \leq 0.05$, $-0.5 \leq x \leq 0.5$, ($i = 1, \dots, 4$)) after 3.50 seconds from the initial position facing downward, and stayed there. These results clearly demonstrate that the proposed controller effectively solves the swing-up control problem of the QIP system.

To visualize the experimental results more clearly, Fig. 7 presents a screenshot from the YouTube video, which demonstrates the results of the experiment and is available at <https://youtu.be/m0ZZFrc811Q> [23] (Video title: Reinforcement learning to achieve real-time control of a quadruple inverted pendulum, Channel name: POSTECH CoCEL).

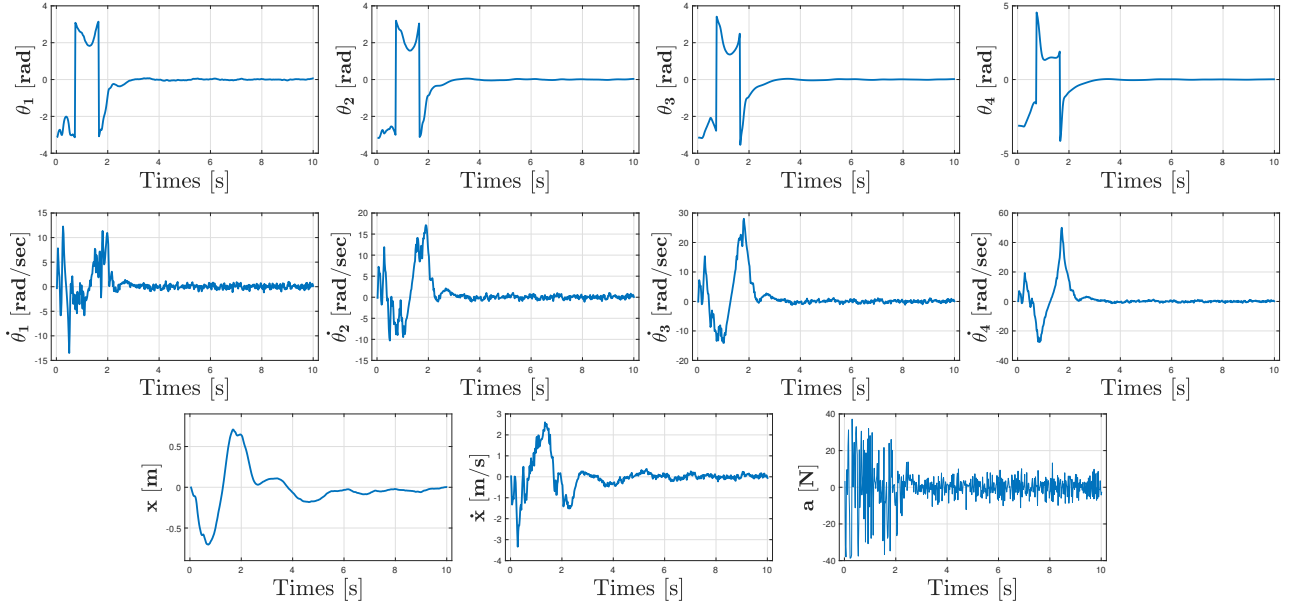


Fig. 5. Trajectories of state variables and action values.

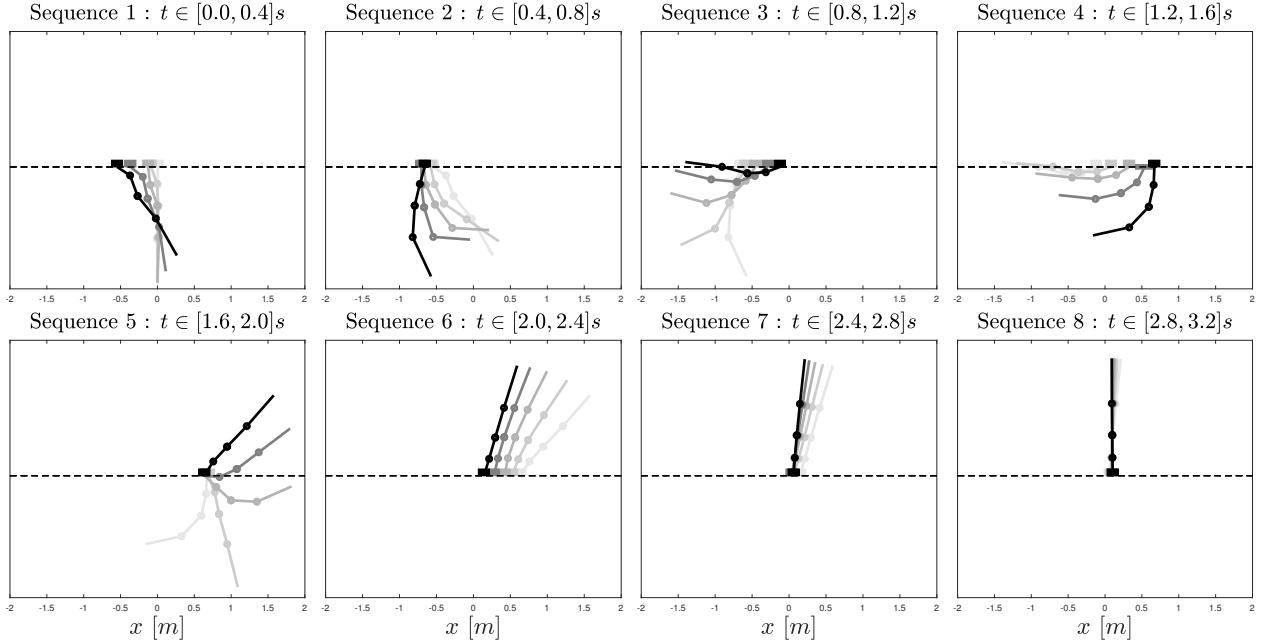


Fig. 6. Snapshots of the swing-up process over time in the QIP system.

5. DISCUSSION

While the preceding sections have focused on training and validating the controller within the nominal model setting, practical deployment inevitably involves variations in system parameters and inherent physical constraints. To explore the feasibility and generalizability of the proposed method beyond nominal conditions, we conducted additional investigations into the robustness of the trained controller and the computational considerations necessary for scaling to more complex systems.

First, we evaluated the controller's robustness under parametric perturbations. Specifically, we modified key physical properties of the QIP system by uniformly scaling all link masses, increasing the lengths of the links, and attaching a small payload to the end of the final link. The trained policy maintained stable swing-up and balance control even when the link masses were increased by up to a factor of two. Furthermore, it remained effective under approximately 4% increases in link lengths and 5% increases in end-effector payload. These results demonstrate that the policy exhibits robust control behavior against

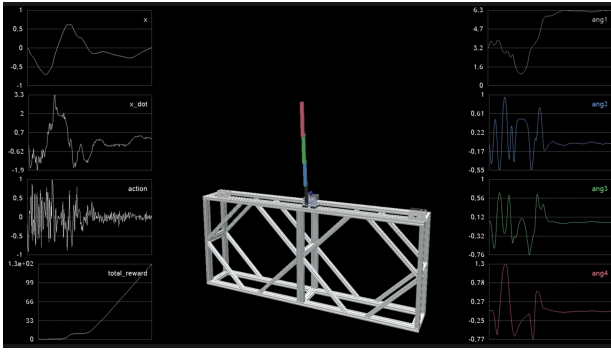


Fig. 7. YouTube video demonstrating the results of the experiment.

moderate variations in physical parameters—even without the use of explicit domain randomization during training. However, for enhanced generalization and real-world applicability, future training regimes should incorporate a broader range of parameter perturbations, including randomized variations in mass distribution and payload.

In parallel, we also considered the computational cost and time complexity of training, particularly in view of extending this approach to n -link pendulum systems. The sampling time (20 ms), episode length (1000 time steps), and total training steps (over 4 million) were not arbitrarily chosen but were determined based on the physical characteristics of real inverted pendulum systems and empirical observations of reinforcement learning convergence. A sampling time of 20 ms was determined based on the 10 ms control interval previously employed in real-time control experiments on physical double- and triple-inverted pendulum (DIP and TIP) systems conducted by the authors' research group. A sampling interval of 10 ms demonstrated stable control performance at the hardware level. Building upon this experience, a longer sampling interval of 20 ms was adopted for the QIP system, in consideration of increased physical parameters such as link length, mass, and moment of inertia. In general, as the system's dynamic response slows down due to greater structural inertia, a longer sampling time becomes permissible. Therefore, the choice of 20 ms reflects a design trade-off that balances the feasibility of real-time control with training efficiency. The episode length of 1000 time steps—equivalent to 20 seconds of simulation—was conservatively chosen to ensure that a full swing-up and balancing sequence could be completed within a single episode. In prior hardware experiments, the swing-up phase typically required 2 to 3 seconds. As this duration may vary with different physical parameter settings, the sampling time and episode length should be jointly designed in accordance with the system's dynamics rather than selected independently. Lastly, the total training duration exceeding 4 million steps does not represent a fixed budget but rather reflects the iterative search for

the highest-performing policy beyond initial convergence. In our experiments, significant reward improvement was observed around 700k steps in the DIP system, 1 million steps in the TIP system, and approximately 1.5 million steps in the QIP system. Continued training through plateau phases allowed us to evaluate multiple candidate policies and select the best among them, which led to the final training length. This progression clearly illustrates that convergence time and computational demands grow substantially with system complexity.

In conclusion, the controller developed in this study has notable robustness against structural perturbations, which is supported by empirically grounded training settings tailored to real-world dynamics. The design parameters used—a sampling time of 20 ms, 1000 time steps episodes, and over 4 million training steps—were not arbitrary, but based on deep understanding of inverted pendulum systems and practical control experience. As the number of links increases, the state dimensionality, learning difficulty, and convergence time grow nonlinearly. Therefore, such design considerations must also be taken into account when scaling to pendulum systems with more than four links, with particular attention to the computational limitations that arise in high-dimensional learning environments.

6. CONCLUSION

For the first time in the world, a closed loop state-feedback controller is successfully designed to achieve the swing-up control of the QIP system in a sample-efficient, data-driven way. To cope with the high complexity of the QIP system, a state-of-the-art TQC algorithm is tactically augmented with data-economical VER. The symmetric structure of the QIP system is fully leveraged to improve sample efficiency and design geometry-aware control. It turns out that the swing-up control problem of the QIP system, which could not be effectively addressed by existing classical control methods or previous RL studies, can be successfully solved.

This research makes a significant contribution to the field of RL-based control for high-dimensional nonlinear systems and establishes a foundation for applying RL algorithms to more complex and diverse systems in the future. Future work will be devoted to applying the proposed RL-based control to an actual QIP system and achieving consistent results.

DECLARATIONS

Conflict of Interest

The authors declare that there is no competing financial interest or personal relationship that could have appeared to influence the work reported in this paper.

Authors' Contributions

Yookyung Oh: Conceptualization, software, formal analysis, investigation, writing—original draft preparation. **Taegun Lee:** Conceptualization, methodology. **Sanghyun Ryoo:** Validation, visualization. **Kyoung Chul Koh:** Supervision. **Soohee Han:** Writing—review and editing, supervision. **Young Sam Lee:** Writing—review and editing, project administration, funding acquisition.

All authors have read and agreed to the published version of the manuscript.

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